Worksheet 2.5 — Radicals & Equivalent Expressions

Grade 11 Functions, University (MCR3U) · Unit 2: Equivalent Algebraic Expressions

The Big Ideas

A radical such as $\sqrt{72}$ is an **exact** number; the calculator's $8.485\ldots$ is only an approximation. We simplify radicals so that equal numbers look equal, and we always check a rewrite on a calculator: *the value must not change*. The two key rules are $\sqrt{a}\sqrt{b} = \sqrt{ab}$ and $\dfrac{\sqrt{a}}{\sqrt{b}} = \sqrt{\dfrac{a}{b}}$. There is **no** rule $\sqrt{a+b} = \sqrt{a} + \sqrt{b}$.

- **Simplify:** divide out the *largest perfect square*: $\sqrt{72} = \sqrt{36 \cdot 2} = 6\sqrt{2}$. Perfect squares: $4, 9, 16, 25, 36, 49, 64, 81, 100, 121, 144, \ldots$
- **Add/subtract** only *like* radicals, after simplifying each: $3\sqrt{5} + 7\sqrt{5} = 10\sqrt{5}$, but $\sqrt{2} + \sqrt{3}$ stays as it is.
- **Multiply:** $(a\sqrt{b})(c\sqrt{d}) = ac\sqrt{bd}$; use FOIL for binomials and remember $(\sqrt{a})^2 = a$.
- **Conjugates:** $(a + \sqrt{b})(a - \sqrt{b}) = a^2 - b$ is rational. To rationalise a binomial denominator multiply top and bottom by the conjugate.
- **Variables:** $\sqrt{x^2} = |x|$; we assume the variables are non-negative to write $\sqrt{18x^3} = 3x\sqrt{2x}$.

Concept Essentials

Simplifying radicals

Divide out the **largest perfect square** factor: $\sqrt{72} = 6\sqrt{2}$. Add or subtract only **like radicals**, after simplifying each. Multiply radicands: $(a\sqrt{b})(c\sqrt{d}) = ac\sqrt{bd}$, and $(\sqrt{a})^2 = a$.

Conjugates and rationalising

$(a+\sqrt{b})(a-\sqrt{b}) = a^2 - b$ is rational. To remove a radical from a denominator multiply top and bottom by $\sqrt{b}$ (monomial) or by the **conjugate** (binomial).

Checking

There is no sum rule: $\sqrt{a+b} \neq \sqrt{a} + \sqrt{b}$. Check any rewrite on a calculator — equivalent exact forms have the same decimal value.

Worked Examples

Example 1 — Simplify a radical

Simplify $\sqrt{200}$.

Solution. The largest perfect square dividing 200 is 100: $\sqrt{200} = \sqrt{100 \cdot 2} = 10\sqrt{2}$.
Check: $\sqrt{200} = 14.142$ and $10\sqrt{2} = 14.142$ ✓.

Example 2 — Mixed and entire radicals

(a) Simplify $\sqrt{63}$. (b) Write $7\sqrt{3}$ as an entire radical.

Solution. (a) $\sqrt{63} = \sqrt{9 \cdot 7} = 3\sqrt{7}$.
(b) Square the coefficient and move it inside: $7\sqrt{3} = \sqrt{49 \cdot 3} = \sqrt{147}$.

Example 3 — Adding and subtracting

Simplify $5\sqrt{12} - 2\sqrt{27} + \sqrt{48}$.

Solution. Simplify first: $\sqrt{12} = 2\sqrt{3}$, $\sqrt{27} = 3\sqrt{3}$, $\sqrt{48} = 4\sqrt{3}$.
$10\sqrt{3} - 6\sqrt{3} + 4\sqrt{3} = 8\sqrt{3}$.

Example 4 — Multiplying

Simplify $(3\sqrt{7})(2\sqrt{14})$.

Solution. $6\sqrt{98} = 6 \cdot 7\sqrt{2} = 42\sqrt{2}$.
Check: $3(2.646) \cdot 2(3.742) = 59.40$ and $42(1.4142) = 59.40$ ✓.

Example 5 — Distributing

Simplify $\sqrt{3}\left(\sqrt{12} + 2\sqrt{5}\right)$.

Solution. $\sqrt{36} + 2\sqrt{15} = 6 + 2\sqrt{15}$.
The terms 6 and $2\sqrt{15}$ are not like terms, so this is final.

Example 6 — FOIL with radicals

Expand $(2\sqrt{3} + 1)(\sqrt{3} - 4)$.

Solution. First $2 \cdot 3 = 6$; Outer $-8\sqrt{3}$; Inner $\sqrt{3}$; Last -4.
$6 - 4 = 2$ and $-8\sqrt{3} + \sqrt{3} = -7\sqrt{3}$: $2 - 7\sqrt{3}$.

Example 7 — Squaring a binomial

Expand $\left(\sqrt{5} - 2\sqrt{3}\right)^2$.

Solution. $a^2 - 2ab + b^2$ with $a = \sqrt{5}$, $b = 2\sqrt{3}$: $5 - 2(\sqrt{5})(2\sqrt{3}) + 4 \cdot 3 = 5 - 4\sqrt{15} + 12$.
$= 17 - 4\sqrt{15}$.

Example 8 — Rationalising denominators

Rationalise (a) $\dfrac{12}{\sqrt{6}}$ (b) $\dfrac{4}{\sqrt{7} - \sqrt{3}}$.

Solution. (a) Multiply by $\dfrac{\sqrt{6}}{\sqrt{6}}$: $\dfrac{12\sqrt{6}}{6} = 2\sqrt{6}$.

(b) Multiply by the conjugate $\dfrac{\sqrt{7} + \sqrt{3}}{\sqrt{7} + \sqrt{3}}$: the denominator is $7 - 3 = 4$, so $\dfrac{4(\sqrt{7} + \sqrt{3})}{4} = \sqrt{7} + \sqrt{3}$.

Example 9 — Radicals with variables

Simplify $\sqrt{72a^5}$ where $a \geq 0$.

Solution. $72 = 36 \cdot 2$ and $a^5 = a^4 \cdot a$, so $\sqrt{36 \cdot a^4 \cdot 2a} = 6a^2\sqrt{2a}$.

Example 10 — Application: distance between points

Find the exact distance between $A(-2, 1)$ and $B(4, 5)$.

Solution. $d = \sqrt{(4-(-2))^2 + (5-1)^2} = \sqrt{36+16} = \sqrt{52}$.
$\sqrt{52} = \sqrt{4 \cdot 13} = 2\sqrt{13} \approx 7.21$ units.

Example 11 — Test before you trust

A student claims $\sqrt{x^2+9} = x+3$. Test it.

x	0	1	2	4
$\sqrt{x^2+9}$	3	3.162	3.606	5
$x+3$	3	4	5	7
same?	✓	×	×	×

Solution. The two sides agree at $x = 0$ only. One mismatch is enough to prove the claim false: there is no sum rule $\sqrt{a+b} = \sqrt{a} + \sqrt{b}$ (here $\sqrt{x^2+9} \neq \sqrt{x^2} + \sqrt{9}$).

Your Turn — Show Your Work

For each question, show all steps clearly in the space provided.

Question 1

Simplify $\sqrt{120}$.

Question 2

Simplify $\sqrt{675}$.

Question 3

Write $4\sqrt{7}$ as an entire radical and check both forms with a calculator.

Question 4

Simplify $4\sqrt{8} + 3\sqrt{18} - \sqrt{50}$.

Question 5

Simplify $(5\sqrt{2})(3\sqrt{6})$.

Question 6

Expand and simplify $\sqrt{5}\left(4\sqrt{10}-3\right)$.

Question 7

Expand and simplify $\left(3\sqrt{2}-4\right)\left(2\sqrt{2}+5\right)$.

Question 8

Expand (a) $\left(\sqrt{11}+\sqrt{5}\right)\left(\sqrt{11}-\sqrt{5}\right)$ (b) $\left(2\sqrt{3}+\sqrt{7}\right)^2$.

Question 9

Rationalise the denominator of $\dfrac{9}{2\sqrt{3}}$.

Question 10

Rationalise the denominator of $\dfrac{6}{4-\sqrt{10}}$ and simplify.

Question 11

Simplify $\sqrt{98x^4y^3}$, where $x, y \geq 0$.

Question 12

A box is 3 m by 4 m by 5 m. Find the exact length of its longest diagonal, $\sqrt{l^2+w^2+h^2}$, and a decimal approximation to 2 decimal places.

Question 13

A student writes $\sqrt{a^2-b^2} = a-b$. Test the claim with $a=5$, $b=3$, then explain which rule the student mis-used.

Question 14 — Challenge

Show that $\sqrt{9+4\sqrt{5}} = 2+\sqrt{5}$ by squaring the right-hand side, and then check both sides with a calculator.

Question Recap & Answer Key

Question Recap

1. Simplify $\sqrt{120}$.
2. Simplify $\sqrt{675}$.
3. Write $4\sqrt{7}$ as an entire radical and check both forms with a calculator.
4. Simplify $4\sqrt{8}+3\sqrt{18}-\sqrt{50}$.
5. Simplify $(5\sqrt{2})(3\sqrt{6})$.
6. Expand and simplify $\sqrt{5}\left(4\sqrt{10}-3\right)$.
7. Expand and simplify $\left(3\sqrt{2}-4\right)\left(2\sqrt{2}+5\right)$.
8. Expand (a) $\left(\sqrt{11}+\sqrt{5}\right)\left(\sqrt{11}-\sqrt{5}\right)$ (b) $\left(2\sqrt{3}+\sqrt{7}\right)^2$.
9. Rationalise the denominator of $\dfrac{9}{2\sqrt{3}}$.
10. Rationalise the denominator of $\dfrac{6}{4-\sqrt{10}}$ and simplify.
11. Simplify $\sqrt{98x^4y^3}$, where $x, y \geq 0$.
12. A box is 3 m by 4 m by 5 m. Find the exact length of its longest diagonal, $\sqrt{l^2+w^2+h^2}$, and a decimal approximation to 2 decimal places.
13. A student writes $\sqrt{a^2-b^2}=a-b$. Test the claim with $a=5$, $b=3$, then explain which rule the student mis-used.
14. Show that $\sqrt{9+4\sqrt{5}}=2+\sqrt{5}$ by squaring the right-hand side, and then check both sides with a calculator.

✓ Answer Key

1. $\sqrt{120}=\sqrt{4\cdot 30}=2\sqrt{30}$.
2. $675=225\cdot 3$, so $15\sqrt{3}$.
3. $4\sqrt{7}=\sqrt{16\cdot 7}=\sqrt{112}$. Both are 10.583.
4. $8\sqrt{2}+9\sqrt{2}-5\sqrt{2}=12\sqrt{2}$.
5. $15\sqrt{12}=15\cdot 2\sqrt{3}=30\sqrt{3}$.
6. $4\sqrt{50}-3\sqrt{5}=20\sqrt{2}-3\sqrt{5}$.
7. $12+15\sqrt{2}-8\sqrt{2}-20=-8+7\sqrt{2}$.
8. (a) $11-5=6$. (b) $12+4\sqrt{21}+7=19+4\sqrt{21}$.
9. $\dfrac{9\sqrt{3}}{2\cdot 3}=\dfrac{3\sqrt{3}}{2}$.
10. Multiply by $\dfrac{4+\sqrt{10}}{4+\sqrt{10}}$: denominator $16-10=6$; $\dfrac{6(4+\sqrt{10})}{6}=4+\sqrt{10}$.
11. $\sqrt{49\cdot 2\cdot x^4\cdot y^2\cdot y}=7x^2y\sqrt{2y}$.
12. $\sqrt{9+16+25}=\sqrt{50}=5\sqrt{2}\approx 7.07$ m.
13. With $a=5$, $b=3$: $\sqrt{25-9}=4$ but $5-3=2$. The student treated the root as if it distributed over subtraction; the root rules apply to *products and quotients* only. ($\sqrt{a^2-b^2}=\sqrt{(a-b)(a+b)}$.)
14. $(2+\sqrt{5})^2=4+4\sqrt{5}+5=9+4\sqrt{5}$ and $2+\sqrt{5}>0$, so $\sqrt{9+4\sqrt{5}}=2+\sqrt{5}$. Calculator: $\sqrt{17.944}=4.236$ and $2+2.236=4.236$ ✓.